



Python Fundamentals: The Route Map

This is your Single Source of Truth.

Ignore the order in YouTube playlists or GFG. Follow THIS order strictly.



Checkpoint 1: The Setup & Basics

Goal: Talking to the computer.

- [] **1. Introduction**
 - What is Python?
 - How to run code (IDLE / VS Code / Online Compiler).
- [] **2. First Program**
 - `print("Hello World")`
 - Handling Syntax Errors (missing quotes, brackets).
- [] **3. Comments**
 - Single line (`#`)
 - Multi-line (`''' ... '''`)
- [] **4. Variables**
 - Creating variables (`x = 10`)
 - Rules for naming variables (no spaces, can't start with numbers).
- [] **5. Data Types**
 - `int` (Integers: 1, 100, -5)
 - `float` (Decimals: 1.5, 3.14)
 - `str` (Strings: "Hello", 'Python')
 - `bool` (Booleans: True, False)
- [] **6. Type Casting (Conversion)**
 - `int()` -> Converts string/float to integer.
 - `str()` -> Converts integer to string.
 - `type()` -> Checking what data type a variable is.
- [] **7. Input**
 - `input()` function.
 - **Crucial:** Understanding that `input()` always takes data as a **String**.



Checkpoint 2: The Math (Operators)

Goal: Performing calculations.

- [] **8. Arithmetic Operators**
 - `+`, `-`, `*`, `/` (Standard math)
 - `//` (**Floor Division:** `5 // 2 = 2`)
 - `%` (**Modulus:** `5 % 2 = 1` -> Remainder)
 - `**` (**Power:** `2 ** 3 = 8`)

- [] **9. Assignment Operators**
 - =, +=, -= (e.g., `x += 1` means `x = x + 1`)
- [] **10. Comparison Operators**
 - == (Equals), != (Not Equals)
 - >, <, >=, <=



Checkpoint 3: The Decision Maker (Logic)

Goal: Making your code smart.

- [] **11. Logical Operators**
 - and (Both must be true)
 - or (At least one is true)
 - not (Reverse the result)
- [] **12. Conditional Statements**
 - if statement.
 - else statement.
 - elif (Else If) statement.
- [] **13. Nested If**
 - Writing an if inside another if.



Checkpoint 4: The Repeater (Loops)

Goal: Doing things 1000 times.

- [] **14. Range Function**
 - `range(5)` -> 0, 1, 2, 3, 4
 - `range(1, 6)` -> 1, 2, 3, 4, 5
 - `range(1, 10, 2)` -> 1, 3, 5, 7, 9 (Step)
- [] **15. For Loop**
 - Looping through a range.
 - Looping through a String ("Python").
- [] **16. While Loop**
 - Repeating while a condition is True.
 - Infinite Loops (How to avoid them).
- [] **17. Loop Controls**
 - `break` (Stop the loop immediately).
 - `continue` (Skip this step, go to next).
 - `pass` (Do nothing - placeholder).



Checkpoint 5: The Containers (Data Structures)

Goal: Storing lists of data.

- [] **18. Lists (The Most Important Topic)**
 - Creating a list: `my_list = [1, 2, 3]`

- Indexing: `my_list[0]`
- Slicing: `my_list[1:3]`
- Modifying: `my_list[0] = 100`
- Methods: `.append()`, `.insert()`, `.remove()`, `.pop()`, `.sort()`, `.reverse()`.
- [] **19. Tuples**
 - Creating a tuple: `my_tuple = (1, 2)`
 - Difference between List and Tuple (Mutable vs Immutable).
- [] **20. Sets**
 - Creating a set: `my_set = {1, 2, 3}`
 - Unique items only (removes duplicates).
- [] **21. Dictionaries**
 - Creating a dict: `my_dict = {'key': 'value'}`
 - Accessing values: `my_dict['key']`
 - Adding/Updating keys.

Checkpoint 6: Text Master (Strings)

Goal: Manipulating text.

- [] **22. String Slicing (Advanced)**
 - Reverse a string: `text[::-1]`
- [] **23. String Methods**
 - `.upper()`, `.lower()`, `.strip()` (Remove spaces).
 - `.replace()`, `.find()`, `.count()`.
 - `.split()` (String to List).
 - `.join()` (List to String).

Checkpoint 7: The Organizer (Functions)

Goal: Writing clean, reusable code.

- [] **24. Defining Functions**
 - `def my_function():`
 - Calling a function.
- [] **25. Parameters & Arguments**
 - Passing data into functions.
- [] **26. Return Statement**
 - Sending data back to the main program.
 - Difference between `print()` and `return`.
- [] **27. Variable Scope**
 - Global Variables vs Local Variables.

 **STOP HERE.**

Congratulations! Once you tick Topic #27, you have officially completed Python Fundamentals. You are now ready to start DSA or Intermediate Python.